

Real-time Object Tracking on the Edge

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Abstract—This paper presents a real-time detection, tracking, counting, and distance estimation framework deployed on the Boston Dynamics Spot robot, equipped with RGB and thermal cameras. Leveraging edge computing devices such as the NVIDIA Jetson Nano, the system autonomously processes data in dynamic terrains with minimal human intervention. A custom-trained YOLOv8 model, fine-tuned on a unique dataset tailored for military applications, is integrated with the StrongSORT algorithm for object tracking. Additionally, a novel geometric calculation methodology enables precise angle estimation and spatial mapping, enhancing situational awareness. The framework's capabilities include live visualization of detected objects, area-based counting, and mapping within an operational environment using ROS-based tools. Field demonstrations conducted at the Army Research Lab validate the system's effectiveness in processing complex thermal and RGB data in real-time. The proposed solution offers significant potential for enhancing autonomous robotic deployments in mission-critical applications.

Index Terms—Real-time Object Detection, Multi-Object Tracking, Computer Vision in Robotics, Kalman Filtering, Edge Computing Optimization.

I. INTRODUCTION

Real-time object detection and tracking are critical capabilities for robotic systems in dynamic environments such as military and security operations. Such systems must efficiently process high volumes of sensor data with minimal latency. Edge computing, leveraging resource-constrained devices like the NVIDIA Jetson Nano, has become increasingly important to achieve autonomous and efficient field deployments.

We present a real-time object detection, tracking, counting, and spatial estimation framework implemented on the Boston Dynamics Spot robot. Our approach employs custom-trained YOLOv8 models optimized for edge deployment, integrated with StrongSORT for tracking, and geometric techniques for spatial awareness. The framework supports visualization and mapping through ROS-based tools, significantly enhancing situational awareness.

Our contributions include: (1) Development of a custom RGB and thermal dataset for military scenarios. (2) Optimization of YOLOv8 for efficient deployment on edge devices. (3) Integration of algorithms for robust tracking and spatial localization. (4) Implementation of real-time visualization for operational support.

Field tests validate system performance, demonstrating its applicability in mission-critical robotic deployments. This pa-

per describes the methodology, experiments, and outcomes, and outlines future research directions.

The rest of the paper is organized as follows: In section II we discuss the more relevant work in the literature, followed by the methodology and field experiments are presented in Section III. Section IV concludes the paper with ideas for future work.

II. RELATED WORK

Object detection and tracking have significantly advanced with deep learning and edge computing technologies. YOLO frameworks introduced by Redmon et al. [1] and subsequent versions like YOLOv4 [2] improved real-time detection accuracy and efficiency. DeepSORT [5] combined appearance-based re-identification with Kalman filtering [11] and the Hungarian algorithm for effective multi-object tracking. Stereo vision-based detection methods [7] combined deep learning with spatial awareness but often faced computational overhead challenges.

CNN architectures, as explored by Szegedy et al. [8], significantly enhanced feature extraction capabilities. However, optimizing these deep learning models for edge deployment remains challenging [10].

Our work addresses these gaps by optimizing detection and tracking through model quantization, memory management, and batch processing for efficient real-time performance on devices like the Jetson Orin Nano. Additionally, real-time visualization and mapping integration provide comprehensive solutions for autonomous robotic applications.

III. METHODOLOGY

Our framework integrates diverse hardware and software components optimized for real-time object detection, tracking, counting, and spatial mapping deployed on the Boston Dynamics Spot robot. The architecture employs RGB and thermal cameras, alongside a VLP-16 LiDAR, interfaced with edge computing devices, specifically NVIDIA Jetson Nano, facilitating autonomous operations.

A. Data Acquisition and Processing

RGB and thermal images are captured using RealSense and ZED cameras, forming a dataset curated explicitly for diverse military scenarios. Images undergo preprocessing such as resizing, normalization, and augmentation to enhance model generalization. High-quality manual annotations provide accurate ground truth data for training and validation.

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B. Model Training

We develop transfer learning techniques by fine-tuning a YOLOv8 model optimized for resource-constrained edge devices. Data augmentation and hyperparameter tuning improve accuracy and computational efficiency. The trained model is converted for deployment with hardware acceleration using TensorRT, significantly improving inference speed and reducing memory usage.

C. Object Tracking

Object tracking employs the StrongSORT algorithm in conjunction with the YOLOv8 model. Each detected object receives a unique identifier to maintain continuity across frames. Kalman filtering enables robust motion prediction, and the Hungarian algorithm efficiently matches detections to object tracks, ensuring accurate tracking even during occlusions and rapid movements.

D. Spatial Awareness: Distance and Angle Estimation

We introduce a geometric method for precise spatial mapping and localization. Distance estimation leverages stereo vision depth calculations based on disparity between camera perspectives. Angle estimation utilizes camera field-of-view data, computing object positions as follows:

$$\theta_{h,v} = \frac{(C_x - W/2)}{W} \times HFOV, \quad (1)$$

$$\phi_{h,v} = \frac{(C_y - H/2)}{H} \times VFOV, \quad (2)$$

where (C_x, C_y) represents the bounding box center, (H, W) image dimensions, and $(HFOV, VFOV)$ denote horizontal and vertical fields-of-view. Objects are mapped accurately onto a 2D environment representation.

E. Object Counting

We employ two counting methods: line-based and area-based. The line-based system tracks objects crossing user-defined virtual boundaries, optimized with Kalman filtering for accuracy. The area-based method counts objects within user-defined polygonal zones, using IoU (Intersection over Union) filtering and persistence tracking techniques to ensure accurate counting and minimize duplicates.

F. Real-Time Visualization and Mapping

Detected objects are visualized in real-time through a dynamically updated 2D maps, leveraging the ROS [12] (Robot Operating System)-based Cartographer package. Continuous position updates enhance situational awareness, and integration with the Command and Control server facilitates remote monitoring, aiding operational decision-making.

G. Field Experiments

To evaluate the performance and robustness of the proposed system, a series of experiments were conducted on legged robots, specifically utilizing the Boston Dynamics Spot robot platform. The objective of these experiments was to assess the system's capabilities in terms of real-time object detection,

tracking, and mapping in dynamic environments with varying terrain conditions.

IV. CONCLUSION AND FUTURE WORK

In this work, we presented a real-time object detection and tracking framework optimized for edge deployment in dynamic field environments. By leveraging state-of-the-art deep learning models such as YOLOv8 and StrongSORT, combined with advanced system optimization techniques, our approach achieves an effective balance between accuracy and efficiency. The integration of real-time visualization and mapping enhances situational awareness, making the system suitable for applications in robotic navigation, surveillance, and assistive systems.

Future work will focus on extending the system's capabilities to include more complex multi-object scenarios and improving robustness under challenging environmental conditions. Enhancements such as adaptive tracking algorithms, self-learning mechanisms, and real-time anomaly detection will be explored to further increase system reliability. Additionally, expanding the framework to support heterogeneous edge computing platforms and cloud-based processing can enable scalability for large-scale deployments. Lastly, further experimentation with alternative sensor modalities, such as radar and acoustic sensors, can provide complementary data to improve detection and tracking performance in diverse operational environments.

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